

INTEGRAL RESEARCH: RAM CALCULUS

Technical Report TR220

Ram Calculus and the Riemann Hypothesis Revisited Certified Local Gram Defects and the Global Positivity Barrier

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Abstract

This report reassesses the Riemann Hypothesis (RH) using the present Ram Calculus corpus, including its corrected operator theory, the Ram Master Transform (RMT), finite and panel Gram operators, weak-atom identities, discrete sampling, microlocal extensions, and resolvent discipline. The immediate stimulus is the August 2026 research draft proving, conditional on the analytic inputs of a companion preprint, that more than 67.3008527% of the nontrivial zeros are simple and lie on the critical line. Its finite seven-gap inequality was independently rerun from the pinned source with Arb interval arithmetic; the committed tests and the full seven-point verifier passed. A second interval run made for this report certified the stronger rational local target 191/50000 and, through the lifting argument proved here, gives the modest further conditional bound 67.3021361950%. This verifies the finite certificates relative to their stated software and floating-point trust base, but it is not an independent proof of all analytic inputs in the companion preprint.

The new argument is naturally a Ram–Weil calculation: windowed exponential atoms generate a positive Gram block, the normalized Gram kernel is an explicit overlap transform, a convex spectral defect retains information lost by rank and two traces, and local interval certificates are lifted to a global zero-density estimate by shifted block pinching. We derive a general local-certificate lifting theorem that applies to any admissible overlap kernel and gives an explicit closed-form density bound. We also prove two limitation results. First, every fixed-width family of bounded local zero statistics is asymptotically insensitive to a zero-density exceptional set; therefore such statistics alone cannot prove RH. Second, dense finite Ram blocks prove RH only if one establishes positivity of *every* block with the continuity and approximation controls required by the Weil criterion. This identifies the exact global positivity barrier.

No proof or disproof of RH is claimed. Earlier Ram reports that treated zeros of a Fourier multiplier as eigenvalues, or inferred compact resolvent from a bounded infinite-dimensional operator, are not used as proofs. Their useful transform identities survive, but their spectral conclusions are replaced by the corrected framework of TR94, TR95, TR97, TR148, TR216, and TR218. The report concludes with a concrete, auditable research program: co-design prime-side-admissible Ram windows and local overlap certificates, enlarge the certified block hierarchy, and seek a uniform coercive limit theorem capable of excluding even a zero-density set of off-line zeros.

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1 Executive Findings and Resolution Status

Status statement 1.1 (Resolution status). As of August 11, 2026, the Riemann Hypothesis remains unsolved. Neither the new 67.3008527% density result nor the analysis in this report proves or disproves RH. The Clay Mathematics Institute continues to list RH as an unsolved Millennium Prize Problem [15].

The investigation produced the following defensible conclusions.

1. **A substantial new density mechanism.** The attached paper [1] refines a rank–trace argument in the companion Claude preprint [3]. It obtains, subject to the correctness of the companion analytic theorem,

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.6730085279277797613 \dots \quad (1)$$

Here $N(T, 2T)$ counts nontrivial zeros with multiplicity and $N_0^s(T, 2T)$ counts simple zeros on the critical line.

2. **Independent finite verification.** The repository was checked out at the pinned commit recorded in Appendix B. All seven unit tests passed. The complete seven-variable interval verifier was rerun with `python-flint` at 128-bit precision. It returned `verified=true` after 707,901 branch-and-bound nodes, maximum depth 37, and the same interval-table hashes recorded by the repository. This is a reproducibility check of the finite inequality, not a substitute for review of the analytic reductions inherited from [3].
3. **A new certified local strengthening.** The verifier constants were changed only from 19/5000 to 191/50000 and from the corresponding pressure cutoff 45,600 to 45,840 grid cells. A fresh exhaustive run returned `verified=true`. The general lifting theorem then gives, under the same analytic hypotheses,

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{13,350,000 H_{\text{MT}} - 26,600}{13,300,149} = 0.6730213619501666 \dots \quad (2)$$

This improves the supplied paper by about 1.2834×10^{-5} in absolute proportion. It is a new computer-assisted corollary conditional on the same companion preprint, not progress from 67% to RH.

4. **A precise Ram Calculus interpretation.** The vectors in the new argument are sampled, normalized windowed-exponential atoms. Their Gram matrix, overlap kernel, finite-block pinching, and defect functional are instances of the RMT and finite Ram operator constructions of TR147, TR165, TR207, TR209, and TR213 [23, 25, 27, 28, 29]. Ram Calculus therefore supplies a useful language and a systematic design space for the method. It does not, merely by relabeling the calculation, add a theorem that forces RH.
5. **A general lifting theorem.** Section 9 proves a kernel-independent theorem. A certified q -point lower bound with constants (α, β) and an analytic baseline H implies the explicit density bound

$$B_m = \frac{mH - C(m-1)}{m - A_m}, \quad C = (q-1)\alpha, \quad A_m = \beta(m - q + 1), \quad (3)$$

for every admissible m with $0 < A_m < 1$. The formula reproduces (1) and exposes exactly which local improvements can produce a better global proportion.

6. **A local-statistics barrier.** Any fixed finite collection of bounded, fixed-width local zero statistics changes by only $o(1)$ after altering $o(N(T))$ zeros. Consequently, no such asymptotic density mechanism can rule out a finite or zero-density family of off-line zeros. This is why a percentage arbitrarily close to 100% still would not by itself prove RH.

7. **The exact remaining Ram target.** A nested dense family of finite Ram–Weil spaces would prove RH if every corresponding Weil Gram matrix were positive semidefinite and the finite-to-infinite approximation were controlled. Conversely, RH implies those finite positive-semidefinite conditions. This is a useful equivalence and computational architecture, but the missing uniform positivity is itself RH-strength.

The best present conclusion is therefore positive but bounded: Ram Calculus aligns exceptionally well with the new finite Gram-defect method and offers a disciplined route to improved zero-density bounds. The corpus does not yet contain the global coercivity theorem needed to pass from density to RH.

2 Claim Discipline and Scope

Long RH investigations become unreliable when exact identities, numerical observations, conditional reductions, and final theorems are mixed. This report uses the following labels.

Label	Meaning in this report
Established	A standard theorem with a conventional proof in the cited literature, such as the functional equation or the Riemann–von Mangoldt formula.
Preprint theorem	A stated theorem in a recent public preprint. It is used with explicit attribution and is not silently promoted to a settled peer-reviewed result.
Certified computation	A finite statement checked by outward-rounded interval arithmetic. Its conclusion is relative to the source, compiler, libraries, hardware semantics, and the correctness of the reduction to the finite problem.
Conditional theorem	A logically proved implication whose hypotheses have not all been established.
Derived theorem	A result proved in this report from clearly stated hypotheses.
Numerical evidence	A floating-point experiment useful for designing a certificate, but not a proof.
Research proposal	A concrete route with a testable milestone and a named obstruction.

The words “Ram–Weil,” “Ram atom,” and “Ram block” refer to a choice of test functions and finite-dimensional representations inside established explicit-formula machinery. They do not assert that the zeta function has been replaced by a different object. Any valid conclusion must survive the translation back to the classical zero and prime sides.

3 The Riemann Hypothesis and the Quantities Being Estimated

For $\operatorname{Re} s > 1$,

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_p (1 - p^{-s})^{-1}. \quad (4)$$

Its meromorphic continuation has a simple pole at $s = 1$. The completed function

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) \quad (5)$$

is entire of order one and satisfies

$$\xi(s) = \xi(1-s), \quad \overline{\xi(s)} = \xi(\bar{s}). \quad (6)$$

The nontrivial zeros therefore occur in the symmetry orbit $\rho, 1 - \rho, \bar{\rho}, 1 - \bar{\rho}$. RH is the statement

$$\zeta(\rho) = 0, \quad 0 < \operatorname{Re} \rho < 1 \quad \implies \quad \operatorname{Re} \rho = \frac{1}{2}. \quad (7)$$

Let $N(T)$ count nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \leq T$, including multiplicity. The Riemann–von Mangoldt formula is

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T). \quad (8)$$

For a dyadic interval $I = (T, 2T]$, write

$$N(T, 2T) = N(2T) - N(T), \quad (9)$$

$$N_0(T, 2T) = \#\{\rho : \gamma \in I, \operatorname{Re} \rho = 1/2\}, \quad (10)$$

$$N_0^s(T, 2T) = \#\{\rho : \gamma \in I, \operatorname{Re} \rho = 1/2, \rho \text{ simple}\}. \quad (11)$$

The last two counts are not interchangeable: a multiple critical-line zero contributes its multiplicity to N but does not contribute to N_0^s .

Two distinctions are decisive.

- A lower bound for N_0/N is a density theorem. RH requires every nontrivial zero to lie on the line, including a possible exceptional set of density zero.
- Simplicity and critical-line location are different assertions. RH does not imply that all zeros are simple, and a simple zero need not be on the critical line.

3.1 The explicit-formula positivity viewpoint

For an admissible test function g , let $\hat{g}$ denote the Mellin/Fourier transform in the normalization appropriate to the explicit formula, and let $\tilde{g}(x) = \overline{\hat{g}(-x)}$. The Weil quadratic form can be written schematically as

$$\mathcal{W}(g, g) = \mathcal{E}(g * \tilde{g}), \quad (12)$$

where $\mathcal{E}$ has equivalent zero-side and prime-side expressions. On the zero side, the functional equation pairs ρ with $1 - \bar{\rho}$; on RH the paired contribution becomes a squared modulus. On the prime side, the same form is a sum of archimedean terms and weighted prime powers. Weil’s criterion states, with the standard admissibility and regularization conditions, that

$$\text{RH} \iff \mathcal{W}(g, g) \geq 0 \quad \text{for every admissible } g. \quad (13)$$

This is the most direct rigorous interface between Ram test atoms and RH. The challenge is the universal quantifier. Positivity for many finite blocks or for a numerically large family of atoms is evidence; positivity on a dense admissible space with a controlled limit is a proof.

4 Literature Position as of August 2026

The classical landscape contains several complementary kinds of progress.

4.1 Critical-line proportions and mollifiers

Hardy’s theorem gives infinitely many critical-line zeros; Selberg proved a positive proportion. Levinson’s mollifier method passed one third, Conrey passed two fifths, and Pratt, Robles, Zaharescu, and Zeindler

proved a value slightly above five twelfths [9]. Such results use delicate mean-value and mollifier estimates and remain density statements.

The August 2026 Claude preprint introduces a different finite-compression and rank–trace framework. It states unconditional lower proportions exceeding two thirds for critical-line and simple critical-line zeros, with optimized constant

$$H_{\text{MT}} = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot\left(\frac{1}{\sqrt{2}}\right) = 0.6725007036794116\dots \quad (14)$$

The official project page reports human mathematical review and a Lean formalization artifact [4, 5]. Because these materials are extremely recent, this report records them as preprint results and makes every downstream conclusion conditional on their analytic inputs.

4.2 Pair correlation and simple zeros

Montgomery’s pair-correlation method, under RH, gives strong information on simple zeros. Baluyot, Goldston, Suriajaya, and Turnage-Butterbaugh developed an unconditional form of the pair-correlation theorem, while the simple-zero consequence in that work assumes a narrow-box or strong zero-density hypothesis [6]. Their later paper studies horizontal distribution and recovers two-thirds conclusions under shrinking-box hypotheses [7]. Goldston and Suriajaya explain why removal of RH from the relevant pair-correlation proof would yield a two-thirds result for zeros both simple and on the line [8].

These works and the Claude preprint share a useful lesson: off-line pairs must be tracked in the linear algebra instead of being silently identified with line zeros. The new Gram-defect refinement exploits still more structure by retaining explicit inner products between the simple-zero vectors.

4.3 Zero density and finite-height verification

Zero-density estimates bound the number of zeros in $\text{Re } s > \sigma$ and feed many prime-distribution applications. Guth and Maynard’s work and the systematic exponent-pair analysis of Tao, Trudgian, and Yang sharpen this technology [11, 12]. These estimates do not show that the count is zero for every height.

Platt and Trudgian rigorously verified RH through height $3 \cdot 10^{12}$ using interval and argument-principle methods [13]. Such a finite-height result is a theorem about a finite region. No amount of finite height alone resolves the universal statement (7).

4.4 Equivalent criteria

Weil positivity, Li coefficients, the Nyman–Beurling closure criterion, de Branges/canonical systems, Jensen-polynomial hyperbolicity, the de Bruijn–Newman flow, and Hilbert–Polya spectral programs are all valuable. An equivalent criterion is not progress toward RH until one proves the criterion’s difficult direction or supplies a quantitatively stronger approximation theorem. This observation governs the Ram corpus audit in Section 8.

5 Audit of the New 67.3008527% Argument

This section reconstructs the logical chain in [1]. Notation is chosen to make its Ram structure explicit.

5.1 Zero decomposition and sampled atoms

Put

$$L = \log \frac{T}{2\pi}, \quad I = (T, 2T], \quad D_0 = T^{1/2}, \quad I' = (T - D_0, 2T + D_0]. \quad (15)$$

The optimized test family inherited from [3] uses a compactly tapered window whose scaled interior tends to

$$t \mapsto \cos(\sqrt{2}, t) \mathbf{1}_{[-1/2, 1/2]}(t). \quad (16)$$

Sample it on a critical-height grid $\{\tau_j\}_{0 \leq j < d}$. For every simple critical-line zero $\rho = 1/2 + i\gamma_\rho$ in I' , define the normalized column vector

$$v_\rho = \frac{1}{\sqrt{aL}} (\varphi(\gamma_\rho - \tau_j))_{0 \leq j < d}, \quad \|v_\rho\| \leq 1. \quad (17)$$

Separate the zeros in I' into simple line zeros S_1 , distinct multiple line zeros S_2 , and unordered off-line pairs P . Write their cardinalities as s_1, s_2, p . Then

$$N(I') \geq s_1 + 2s_2 + 2p. \quad (18)$$

Let V have the vectors (17) as columns and set

$$P_1 = VV^*, \quad M = V^*V. \quad (19)$$

Thus P_1 is the simple-line-zero contribution to the sampled operator and M is its Gram matrix. If $\hat{A}$ denotes the full compressed Hermitian matrix and $Q_0 = \hat{A} - P_1$, the companion paper's inertia decomposition gives

$$n_+(Q_0) \leq s_2 + p. \quad (20)$$

5.2 The convex spectral defect

Define

$$\Psi(t) = \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2, \end{cases} \quad D(M) = \text{tr } \Psi(M). \quad (21)$$

The function is convex and nonnegative. It vanishes only at $t = 1$, so $D(M)$ measures the failure of the simple-zero atom family to be an orthonormal system.

Lemma 5.1 (Stability-enhanced rank–trace inequality). *Let V be a $d \times r$ matrix whose columns have norm at most one. Put $P = VV^* \succeq 0$ and $M = V^*V$. If Q is Hermitian and $n_+(Q) \leq b$, then*

$$\|P + Q\|_F^2 \geq 4 \text{tr}(P + Q) - 3r - 4b + D(M). \quad (22)$$

Proof. Write $Q = Q_+ - Q_-$ with $Q_\pm \succeq 0$, orthogonal supports, and $\text{rank } Q_+ \leq b$. If q_j are the positive eigenvalues of Q_+ , then $q_j^2 \geq 4q_j - 4$, hence

$$\|Q_+\|_F^2 \geq 4 \text{tr } Q_+ - 4b. \quad (23)$$

Let $p_1 \geq \dots \geq p_r \geq 0$ be the eigenvalues of M , padded by zeros, and let $n_1 \geq n_2 \geq \dots \geq 0$ be those of Q_- . Von Neumann's trace inequality gives

$$\|P - Q_-\|_F^2 + 4 \text{tr } Q_- \geq \sum_{j \leq r} ((p_j - n_j)^2 + 4n_j). \quad (24)$$

Direct minimization over $n \geq 0$ yields

$$\min_{n \geq 0} ((p - n)^2 + 4n) = 2p - 1 + \Psi(p). \quad (25)$$

Consequently,

$$\|P - Q_-\|_F^2 \geq 2 \operatorname{tr} P - r + D(M) - 4 \operatorname{tr} Q_-. \quad (26)$$

Combine this with (23), use the orthogonality of Q_+ and Q_- , and finally use $\operatorname{tr} P \leq r$. $\square$

Applying Lemma 5.1 to $P_1 + Q_0 = \hat{A}$, using (18)–(20), and importing the prime-side trace estimates of [3] gives the baseline with a retained central Gram block M° :

$$N_0^s(T, 2T) \geq H_{\text{MT}} N(T, 2T) + D(M^\circ) - o(N(T, 2T)). \quad (27)$$

The original rank–trace proof drops $D(M^\circ) \geq 0$. The extension keeps and estimates it.

5.3 The overlap kernel

Normalize ordinates by the mean-spacing scale

$$h = \frac{2\pi}{L}, \quad x_\rho = \frac{\gamma_\rho - T}{h}. \quad (28)$$

After deleting $o(N(T, 2T))$ endpoint zeros, the atom overlap for every fixed normalized separation is

$$\langle v_\rho, v_{\rho'} \rangle = k(x_\rho - x_{\rho'}) + o(1), \quad (29)$$

uniformly on compact separation ranges, where

$$k(x) = \frac{K(x)}{K(0)}, \quad w(x) = k(x)^2, \quad (30)$$

$$K(x) = \int_{-1/2}^{1/2} \cos(\sqrt{2}, t) \cos(2\pi x t) dt \quad (31)$$

$$= \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}}, \quad (32)$$

$$K(0) = \sqrt{2} \sin(1/\sqrt{2}). \quad (33)$$

The apparent singularities in the closed form are removable. Formula (31) is precisely a normalized autocorrelation transform of the limiting window.

5.4 The certified seven-gap inequality

For six nonnegative consecutive gaps $g_1, \dots, g_6$, define

$$F_6(g_1, \dots, g_6) = \frac{1}{3000} \sum_{i=1}^6 g_i + \sum_{r=1}^6 \frac{2}{7-r} \sum_{i=1}^{7-r} w(g_i + \dots + g_{i+r-1}). \quad (34)$$

The twenty-one kernel terms are exactly the pairwise separations among seven ordered points. The computer-assisted proposition is

$$F_6(g_1, \dots, g_6) \geq \frac{19}{5000} \quad \text{for all } g_i \geq 0. \quad (35)$$

The verifier first uses the pressure term to remove $\sum g_i \geq 11.4$. It covers the remaining compact set by grid cells of width $1/4000$, encloses w and a second derivative lower bound with Arb, and prunes boxes using interval lower bounds, convex tangents, or subdivision. An unresolved terminal box causes failure; thus a successful traversal is a proof of the finite inequality relative to the trust base.

5.5 From seven points to a long block

Let $y_1 < \dots < y_m$ and

$$E_m = 2 \sum_{1 \leq i < j \leq m} w(y_j - y_i). \quad (36)$$

Summing (35) over consecutive seven-point windows gives

$$E_m + \frac{1}{500}(y_m - y_1) \geq \frac{19}{5000}(m - 6). \quad (37)$$

Indeed, a pair spanning r gaps occurs at most $7 - r$ times and therefore receives total weight at most two. A single gap occurs in at most six windows, giving the span coefficient $6/3000 = 1/500$.

For any positive semidefinite Hermitian G ,

$$D(G) \geq \min \left\{ 1, 2 \sum_{i < j} |G_{ij}|^2 \right\}. \quad (38)$$

If all eigenvalues are at most two, this follows from $D(G) = \text{tr}(G - I)^2$; if one eigenvalue exceeds two, its contribution to $D(G)$ exceeds one.

Choose $m = 269$. Then

$$A_0 = \frac{19}{5000}(m - 6) = \frac{4997}{5000} < 1. \quad (39)$$

Equations (29), (37), and (38) imply, uniformly for every consecutive retained block B of 269 simple zeros,

$$D(G_B) + \frac{1}{500} \text{span}(B) \geq A_0 - o(1). \quad (40)$$

5.6 Shifted block pinching and the final constant

For each of the m offsets, partition the ordered simple zeros into full consecutive blocks of size m . Convexity and unitary invariance of $G \mapsto \text{tr} \Psi(G)$ imply that pinching cannot increase the amount of defect available on the right side; equivalently,

$$D(M^\circ) \geq \sum_B D(G_B). \quad (41)$$

Averaging over offsets, each normalized gap is charged to at most $m - 1$ block spans. The Riemann–von Mangoldt formula controls the total normalized span by $N(T, 2T) + o(N(T, 2T))$. Therefore

$$D(M^\circ) \geq \frac{4997}{1,345,000} N_0^s(T, 2T) - \frac{268}{134,500} N(T, 2T) - o(N(T, 2T)). \quad (42)$$

Substitution into (27) yields

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{1,345,000 H_{\text{MT}} - 2,680}{1,340,003}, \quad (43)$$

which is (1).

6 Independent Reproducibility Audit

6.1 Pinned artifacts

The supplied PDF was compared with the paper in the pinned repository. The repository contains the paper source, tests, verifier source, and recorded certificates. The audited commit was

040c5e899e65...fe10761d.

The repository labels the work a research draft generated by GPT-5.6 Sol and invites independent verification and peer review [2]. The PDF itself does not provide a conventional personal author name; the repository’s `CITATION.cff` identifies the project author as `ainta`. This report preserves that attribution rather than inventing one.

6.2 Executed checks

The following checks were executed locally.

Check	Result
Repository pin	Exact requested commit checked out.
Package installation	Editable install completed with <code>python-flint 0.9.0</code> .
Unit tests	Seven tests executed; seven passed.
Official seven-point verifier	<code>verified=true</code> .
Precision and grid	128 bits; mesh 1/4000.
Search	707,901 visited nodes, 354,315 pruned nodes, 353,586 splits, maximum depth 37.
Kernel-table hash	<code>a9992300d2bf...ea55a2c</code> ; full value in Appendix B.
Second-derivative-table hash	<code>7913c5511a57...e83570a</code> ; full value in Appendix B.
Stability lemma spot check	20,000 random finite matrix cases; no violation observed. This is a debugging check, not part of the proof.
TR220 strengthened target	$F_6 \geq 191/50000$ returned <code>verified=true</code> : 786,215 nodes, 393,472 pruned, 392,743 splits, maximum depth 43, and pressure cutoff 45,840.
Strengthened kernel-table hash	<code>f79a9147ffc3...23838571</code> ; full value in Appendix B.
Strengthened second-derivative hash	<code>cc98102590db...35e5e3f5</code> ; full value in Appendix B.

The successful rerun confirms that the distributed code proves (35) under its implementation assumptions. It does not certify the 35-page analytic companion preprint. The companion Lean repository reports a sorry-free formalization of its main theorems; a full local Lean rebuild was not attempted because its dependency/build footprint is multi-gigabyte. These are distinct trust statements and should remain distinct.

6.3 Trust boundary

A computer-assisted proof of the present kind depends on four layers:

1. the mathematical reduction from zeta zeros to the finite inequality;

2. the exact specification of the finite inequality and all constants;
3. the correctness of interval formulas, directed rounding, and search coverage;
4. the software/hardware trust base, including Python, `python-flint`, Arb, the compiler, and machine arithmetic.

The local audit inspected the interval table construction, directed rounding, range minima, convex-tangent pruning, pressure pruning, terminal-cell failure condition, and certificate hashes. No implementation defect was found. A future archival release should additionally provide a small independent verifier in a second language or a proof-assistant import of the finite certificate.

7 Ram Calculus Interpretation

7.1 RMT atoms and the Gram operator

The Ram Master Transform can be written abstractly as an analysis map

$$(\mathcal{R}_\theta f)(\lambda) = \langle f, a_{\theta, \lambda} \rangle, \quad (44)$$

where $a_{\theta, \lambda}$ is a parameterized atom: translated, modulated, scaled, differentiated, chirped, or otherwise adapted to the operator and data. For a finite atom family $a_1, \dots, a_d$, the analysis matrix and Gram matrix are

$$V = (v_1 \ \cdots \ v_r), \quad G = V^* V, \quad G_{jk} = \langle v_j, v_k \rangle. \quad (45)$$

The vectors (17) are a real Gabor-type RMT family: the zero ordinate translates a fixed window relative to a uniform sampling grid. Their limiting normalized autocorrelation is k , and the squared coherence is $w = k^2$.

Thus the new proof is not merely analogous to Ram Calculus. Its finite mechanism is a direct instance of the following Ram pipeline:

$$\begin{aligned} \text{admissible window} &\longrightarrow \text{sampled atoms} \longrightarrow \text{Gram kernel} \longrightarrow \text{spectral defect} \\ &\longrightarrow \text{certified local block} \longrightarrow \text{global pinching bound.} \end{aligned} \quad (46)$$

7.2 What each corpus component contributes

Ram component	Role in the present RH program
RMT/Gabor atoms (TR147)	Supplies the parameterized windowed-exponential analysis family and its overlap/autocorrelation interpretation.
Analytic core plus residual (TR165)	Separates exact explicit-formula and Gram structure from learned or numerical residual search; only the analytic core may carry a proof.
Weak atom equations (TR207)	Moves derivatives and oscillations onto smooth test atoms and requires calibration, domain, and noise assumptions to remain visible.
Panel and block operators (TR209, TR215)	Combines several admissible windows or scales without confusing each panel's normalization and inertia.
Discrete sampling (TR213)	Tracks quadrature, truncation, frame bounds, and finite-matrix conditioning.

Ram component	Role in the present RH program
Microlocal and pseudodifferential extension (TR216)	Provides a disciplined language for localized symbol probes, while explicitly separating exact differential identities from asymptotic wave-packet claims.
Resolvents and closed operators (TR218)	Prevents finite Gram matrices or bounded multipliers from being mistaken for a self-adjoint operator with compact resolvent.

7.3 What Ram Calculus does not supply automatically

The following implications are invalid without additional theorems:

transform zero $\nRightarrow L^2$ eigenfunction,
 real scalar symbol $\nRightarrow$ self-adjoint differential realization,
 finite Hermitian matrices $\nRightarrow$ self-adjoint limiting operator,
 many PSD blocks $\nRightarrow$ universal Weil positivity,
 density $1 - o(1) \nRightarrow$ RH.

Ram Calculus helps design and organize test functions, compute kernels, expose finite moments, and certify local inequalities. The arithmetic trace estimates and the global positivity limit remain independent proof obligations.

8 Audit of Earlier Ram RH Reports

The corpus was searched for RH, GRH, Hilbert–Polya, zeta, Weil, Li, Nyman–Beurling, de Branges, explicit-formula, zero-density, and spectral determinant discussions. The reports most relevant to the present analysis are summarized below.

Report	Useful surviving content	Current status for RH
TR20, TR23, TR65	FOIRT/Ram transform identities, symmetry bookkeeping, and attempts to encode ξ in a multiplier.	The claimed Hilbert–Polya closure is not valid. Isolated zeros of a multiplier do not normally give normalizable point spectrum, and the stated compact-resolvent step fails.
TR73, TR96	Adelic and arithmetic multiscale organization; possible prime/place panels.	A research architecture, not an established trace formula proving RH.
TR94	Five corrected paths: Weil positivity, de Branges, Li/Nyman–Beurling, Jensen/de Bruijn–Newman, and adelic trace formulas.	Correctly identifies the missing positivity theorem in every path.
TR95, TR97	GRH extensions and a corrected conditional spectral formulation.	Conditional only; the self-adjoint realization and exact spectral determinant remain assumptions.
TR147	Comprehensive RMT, local kernels, finite jets, and wave-packet families.	Directly useful for the finite Gram-defect method; it does not supply the prime-side estimate.

Report	Useful surviving content	Current status for RH
TR148	Spectral/zero-density reassessment and explicit retraction of the overstated TR20/TR65 conclusion.	Adopted as the governing RH status in this report.
TR165	Analytic core plus learned residual architecture.	Numerics may search for windows or certificates; a learned residual is not proof unless converted to exact inequalities.
TR184	Roadmap for general Ram operators and extensions.	Useful project organization, with RH-specific proof obligations still open.
TR207– TR219	Robust calibration, multidimensional and panel operators, discrete sampling, nonlinear/coupled systems, microlocal analysis, fractional order, and resolvent/boundary theory.	Strengthens the rigor of the computational framework; does not create a missing RH positivity theorem.

8.1 Correction ledger

The following corrections are binding for TR220.

1. **Multiplier zeros.** If a translation-invariant operator acts as multiplication by $m(\xi)$ in Fourier space, an isolated zero of m normally does not produce a nonzero L^2 function supported at that point. A delta distribution is not an L^2 eigenvector.
2. **Compact resolvent.** A bounded operator on an infinite-dimensional Hilbert space cannot have compact resolvent at a point of its resolvent set: if $(A - zI)^{-1}$ were compact, then $I = (A - zI)(A - zI)^{-1}$ would be compact.
3. **Weighted adjoints.** Introducing a Gaussian or other weight changes the adjoint and the operator domain. Formal integration by parts does not establish self-adjointness, essential self-adjointness, or confinement.
4. **Finite sections.** Hermitian finite sections can converge spectrally in misleading ways unless one proves domain convergence, resolvent convergence, exclusion of spectral pollution, and the exact determinant identity.
5. **Equivalent criteria.** Rewriting Weil or Li positivity in Ram coordinates is valuable, but positivity in the new coordinates is precisely the hard part.
6. **Computation.** Numerical agreement at many zeros, large matrix sizes, or high heights cannot rule out a later off-line zero. Certified finite computations must be stated with their finite scope.

The correction does not discard the earlier work. It retains the test families, kernels, moment identities, multiresolution organization, and algorithmic diagnostics, while removing conclusions that did not follow.

9 A General Local-Certificate Lifting Theorem

This section abstracts the new paper’s seven-point calculation. The theorem is useful beyond the particular cosine window and separates the analytic number theory input from the finite Ram certificate.

9.1 Hypotheses

Let $S_T = N_0^s(T, 2T)$ and $N_T = N(T, 2T)$. Suppose an analytic argument gives

$$S_T \geq HN_T + D_T - o(N_T), \quad (47)$$

where $H \in [0, 1]$ and $D_T \geq 0$ is a convex Gram-defect functional on the retained simple-zero vectors. Assume the retained family has $S_T - o(N_T)$ vectors.

Assume the retained normalized ordinates have total span at most $N_T + o(N_T)$, their normalized overlap converges uniformly on fixed compact sets to a real kernel k , and put $w = k^2$. Assume D_T is additive under direct sums and has the pinching property

$$D_T \geq \sum_B D(G_B) \quad (48)$$

for every partition into principal Gram blocks. Finally, assume every positive semidefinite block G satisfies

$$D(G) \geq \min \left\{ 1, 2 \sum_{i < j} |G_{ij}|^2 \right\}. \quad (49)$$

Finally, suppose there are $q \geq 2$, $\alpha > 0$, and $\beta > 0$ such that every $g_1, \dots, g_{q-1} \geq 0$ obeys

$$\alpha \sum_{i=1}^{q-1} g_i + \sum_{r=1}^{q-1} \frac{2}{q-r} \sum_{i=1}^{q-r} w(g_i + \dots + g_{i+r-1}) \geq \beta. \quad (50)$$

Theorem 9.1 (Local-certificate lifting). *Under the preceding hypotheses, set*

$$C = (q-1)\alpha, \quad A_m = \beta(m-q+1). \quad (51)$$

For every integer $m \geq q$ satisfying $0 < A_m < 1$,

$$\liminf_{T \rightarrow \infty} \frac{S_T}{N_T} \geq B_m := \frac{mH - C(m-1)}{m - A_m}. \quad (52)$$

The bound improves the baseline H exactly when

$$HA_m > C(m-1). \quad (53)$$

Proof. Let $y_1 < \dots < y_m$ be a block of normalized simple-zero ordinates and let

$$E_m = 2 \sum_{1 \leq i < j \leq m} w(y_j - y_i). \quad (54)$$

Sum (50) over the $m - q + 1$ consecutive q -point windows. A pair spanning r consecutive gaps occurs in at most $q - r$ windows, so its total coefficient is at most two. Each individual gap occurs in at most $q - 1$ windows. Hence

$$E_m + C(y_m - y_1) \geq A_m. \quad (55)$$

If $C(y_m - y_1) \geq A_m$, the desired block inequality is immediate from $D(G) \geq 0$. Otherwise the span is bounded by the fixed number A_m/C . Uniform overlap convergence and (49), together with $A_m < 1$, give

$$D(G_B) + C \text{span}(B) \geq A_m - o(1) \quad (56)$$

uniformly over all retained blocks.

Partition the ordered points into full blocks of size m for each of the m cyclic offsets. Convex pinching gives $D_T \geq \sum_B D(G_B)$ for each partition. Average (56) over the offsets. There are $S_T/m + o(N_T)$ full blocks on average; every consecutive gap occurs in at most $m - 1$ spans. The normalized total span is at most $N_T + o(N_T)$. Therefore

$$D_T \geq \frac{A_m}{m} S_T - \frac{C(m-1)}{m} N_T - o(N_T). \quad (57)$$

Substitute (57) into (47) and rearrange:

$$\left(1 - \frac{A_m}{m}\right) S_T \geq \left(H - \frac{C(m-1)}{m}\right) N_T - o(N_T). \quad (58)$$

Division by N_T and passage to the lower limit prove (52). Direct subtraction gives

$$B_m - H = \frac{H A_m - C(m-1)}{m - A_m}, \quad (59)$$

which proves (53). $\square$

Corollary 9.2 (Block-size optimization). *Write $r = q - 1$, so $A_m = \beta(m - r)$. If m is temporarily treated as a real variable, then the sign of $\frac{d}{dm} B_m$ is the sign of*

$$\beta r(H - C) - C(1 - \beta). \quad (60)$$

If this expression is positive, the best integer allowed by $A_m < 1$ is the largest integer satisfying

$$m < r + \frac{1}{\beta}. \quad (61)$$

Proof. Use

$$B_m = \frac{m(H - C) + C}{m(1 - \beta) + \beta r} \quad (62)$$

and differentiate. The m -dependent terms cancel, leaving (60) divided by a positive square. $\square$

Corollary 9.3 (Recovery of the supplied paper). *Take*

$$q = 7, \quad \alpha = \frac{1}{3000}, \quad \beta = \frac{19}{5000}, \quad C = \frac{1}{500}, \quad H = H_{\text{MT}}. \quad (63)$$

Expression (60) is positive, and the largest admissible integer is $m = 269$. Theorem 9.1 gives

$$B_{269} = \frac{1,345,000 H_{\text{MT}} - 2,680}{1,340,003} = 0.6730085279277797613 \dots \quad (64)$$

Corollary 9.4 (TR220 certified strengthening). *The interval run recorded in Appendix B certifies*

$$F_6(g_1, \dots, g_6) \geq \frac{191}{50000} \quad (g_i \geq 0). \quad (65)$$

Take $q = 7$, $\alpha = 1/3000$, $\beta = 191/50000$, and $C = 1/500$. Expression (60) is positive, and the largest integer with $A_m < 1$ is $m = 267$. Since

$$A_{267} = \frac{191}{50000}(267 - 6) = \frac{49851}{50000}, \quad (66)$$

Theorem 9.1 yields the bound (2).

Remark 9.5 (Scope of the strengthening). Only the finite target and its logically required pressure cutoff were changed. The kernel, coefficients, grid, interval formulas, precision, branching rules, and failure policy were unchanged. This is enough to prove (65) relative to the verifier's trust base. The corollary still imports (27) from [3]; it is therefore conditional on that recent analytic theorem.

9.2 Interpretation of the formula

The formula separates three resources.

- H is analytic number theory: trace estimates, support restrictions, and treatment of off-line pairs.
- β is local geometry: the unavoidable coherence defect among q consecutive atoms.
- C is transport cost: the price paid when local blocks are shifted and their spans are charged to the total interval.

A better kernel certificate is useful only if $H\beta(m - q + 1)$ beats the span cost $C(m - 1)$. This prevents blind optimization of a local minimum that has no global effect.

10 Why Fixed Local Statistics Cannot Prove RH

The new argument is powerful because it improves a positive proportion. Its locality also imposes a sharp logical limit.

Theorem 10.1 (Invisibility of a zero-density edit). *For each T , let $X_T = (x_1 < \dots < x_{n_T})$ be an ordered point sequence with $n_T \rightarrow \infty$. Fix q and a bounded function $f : \mathbb{R}^{q-1} \rightarrow \mathbb{R}$, $|f| \leq M$. Define the normalized fixed-width local statistic*

$$\mathcal{L}_T(X_T) = \frac{1}{n_T} \sum_{j=1}^{n_T-q+1} f(x_{j+1} - x_j, \dots, x_{j+q-1} - x_{j+q-2}). \quad (67)$$

If Y_T is obtained from X_T by inserting, deleting, or changing $e_T = o(n_T)$ points, then

$$\mathcal{L}_T(Y_T) - \mathcal{L}_T(X_T) = o(1). \quad (68)$$

The conclusion remains true for any fixed finite vector of such statistics.

Proof. One point edit can affect only $O(q)$ consecutive q -point windows, apart from an $O(1)$ change in the normalization. Hence e_T edits change the unnormalized sum by at most $O(Mqe_T)$ and the normalization by $O(e_T/n_T)$. Division by n_T gives $O(Mqe_T/n_T) = o(1)$. A fixed finite vector follows componentwise. $\square$

Corollary 10.2 (Density methods do not exclude exceptional zeros). *Any argument whose decisive asymptotic input is a fixed finite family of bounded, fixed-width local zero statistics cannot distinguish all-on-line configurations from configurations containing $o(N(T))$ exceptional off-line zeros. Additional nonlocal, height-uniform, or coercive information is required to prove RH.*

Remark 10.3. The certificate (50) contains a linear span term and is not literally bounded. In the block-lifting argument its total contribution is controlled by the global normalized span. The same obstruction remains: changing $o(N_T)$ zeros changes only $o(N_T)$ local blocks and an $o(N_T)$ amount of controlled span. Therefore the resulting normalized density bound is insensitive to a zero-density exception.

This theorem formalizes a point made in the companion preprint: its trace inputs are compatible with $o(N)$ off-line zeros. It also explains why larger fixed blocks, more accurate local kernels, or a proportion approaching one do not automatically cross the RH threshold.

11 The Exhaustive Ram–Weil Positivity Bridge

The local barrier does not make Ram Calculus irrelevant to RH. It tells us that the endpoint must be a global positivity or coercivity theorem rather than only another fixed-width density statistic.

Theorem 11.1 (Dense finite Ram–Weil criterion). *Let $\mathcal{A}$ be an admissible topological vector space of Weil test functions, and let $\mathcal{W} : \mathcal{A} \times \mathcal{A} \rightarrow \mathbb{C}$ be the continuous Hermitian Weil form. Let*

$$\mathcal{A}_1 \subset \mathcal{A}_2 \subset \cdots \subset \mathcal{A}, \quad \overline{\bigcup_{n \geq 1} \mathcal{A}_n} = \mathcal{A}, \quad (69)$$

where each $\mathcal{A}_n$ is spanned by finitely many Ram atoms $a_{n,1}, \dots, a_{n,d_n}$. Define

$$W_n = (\mathcal{W}(a_{n,j}, a_{n,k}))_{j,k=1}^{d_n}. \quad (70)$$

Then

$$W_n \succeq 0 \text{ for every } n \iff \mathcal{W}(g, g) \geq 0 \text{ for every } g \in \mathcal{A}. \quad (71)$$

With the standard Weil criterion, either side is equivalent to RH.

Proof. If the full form is nonnegative, each finite restriction is nonnegative. For the converse, $W_n \succeq 0$ implies $\mathcal{W}(g, g) \geq 0$ for every $g \in \mathcal{A}_n$. This holds on the union. For arbitrary $g \in \mathcal{A}$, choose g_j in the union with $g_j \rightarrow g$. Continuity of the Hermitian form gives $\mathcal{W}(g, g) = \lim_j \mathcal{W}(g_j, g_j) \geq 0$. $\square$

Criterion 11.2 (A quantitative sufficient form). *Suppose $P_n : \mathcal{A} \rightarrow \mathcal{A}_n$ are approximants and there are seminorms $\|\cdot\|_{\mathcal{A}}$, a constant K , and errors $\varepsilon_n \downarrow 0$ such that $\|P_n g\|_{\mathcal{A}} \leq K \|g\|_{\mathcal{A}}$ and*

$$|\mathcal{W}(g, g) - \mathcal{W}(P_n g, P_n g)| \leq \varepsilon_n \|g\|_{\mathcal{A}}^2. \quad (72)$$

If certified finite matrices satisfy, for $u \in \mathcal{A}_n$,

$$\mathcal{W}(u, u) \geq -\delta_n \|u\|_{\mathcal{A}}^2, \quad \delta_n \downarrow 0, \quad (73)$$

then $\mathcal{W}(g, g) \geq 0$ for all g and RH follows.

Indeed,

$$\mathcal{W}(g, g) \geq -(\varepsilon_n + K^2 \delta_n) \|g\|_{\mathcal{A}}^2, \quad (74)$$

and the right side tends to zero. Exact positive semidefiniteness is the special case $\delta_n = 0$.

Criterion 11.2 displays the missing ingredients in algorithmic form: a dense admissible Ram system, explicit approximation error, and a finite lower spectral bound that dominates the error uniformly. Computing many positive matrices without (72) or a matching margin does not establish the criterion.

11.1 Why trace and inertia are insufficient

The first two traces constrain

$$\sum_j \lambda_j, \quad \sum_j \lambda_j^2, \quad (75)$$

and an inertia estimate constrains the number of positive or negative directions. None of these alone proves $\lambda_{\min} \geq 0$. The convex defect $D(M)$ adds information about the distribution of eigenvalues and improves a density count, but it still permits a small exceptional negative subspace. RH requires eliminating every negative direction in every admissible limit.

12 Can the New Result Be Developed Further?

The answer is yes for density bounds and method design, but not yet for RH. Theorem 9.1 makes several extensions precise.

12.1 Optimize the local certificate and block size together

For fixed analytic baseline H and kernel w , solve

$$\max_{q, \alpha, \beta, m} \frac{mH - (q-1)\alpha(m-1)}{m - \beta(m-q+1)} \quad (76)$$

subject to:

1. the global certificate (50) for all nonnegative gaps;
2. $0 < \beta(m-q+1) < 1$;
3. the compact-separation range required by uniform overlap convergence;
4. prime-side support and trace constraints for the chosen window.

This avoids optimizing β in isolation. A larger β can force a smaller admissible m , and a larger pressure coefficient α raises the global span cost.

12.2 Co-design rather than post-process the window

The cosine window in [3] was optimized for the baseline trace constant H_{MT} . The extension shows that its autocorrelation geometry also matters. A better objective is therefore

$$\theta \mapsto B_m(H(\theta), \alpha(\theta), \beta_q(\theta)), \quad (77)$$

where θ parameterizes an admissible Ram window. The optimization must retain the Fourier support and regularity needed by the explicit formula. An attractive kernel with no provable prime-side trace estimate is unusable.

12.3 Multiple panels

TR209 and TR215 suggest several windows $k_1, \dots, k_R$ and the direct-sum or weighted Gram block

$$G^{\text{pan}} = \sum_{r=1}^R \omega_r G^{(r)}, \quad \omega_r \geq 0. \quad (78)$$

If each panel has an admissible prime-side estimate, a joint local certificate may prevent simultaneous near-orthogonality more strongly than any one kernel. The rigorous problem is to propagate the panel weights through the rank-trace and inertia steps without double-counting zeros or prime terms.

12.4 Higher spectral defects

The function Ψ is tailored to the scalar minimization in Lemma 5.1. Higher trace information could support a sharper convex function Ψ_r satisfying a matrix inequality of the form

$$\|P + Q\|_F^2 + \sum_{j=3}^r c_j \text{tr}(P + Q)^j \geq \text{counting terms} + \text{tr } \Psi_r(V^*V). \quad (79)$$

This is only useful if the corresponding prime-side moments can be estimated with errors $o(N_T)$. Higher moments generally require longer arithmetic support and are not a free linear-algebra improvement.

12.5 From local to mesoscopic scales

The local-invisibility theorem suggests allowing the block width $q = q(T)$ to grow. A plausible mesoscopic program would require all of the following:

1. overlap convergence uniform for separations increasing with T ;
2. certificates whose dimension grows without exponential loss;
3. prime-side trace estimates uniform in the increasing support;
4. a lower bound that remains coercive after normalization;
5. control of endpoints, multiplicities, and off-line pairs uniformly in the growing scale.

No such package is currently available. Nevertheless, it is more aligned with the global nature of RH than indefinitely improving a fixed q .

13 A Ram–Weil Research Algorithm

Algorithm 13.1 (Auditable window-to-density pipeline). For a parameterized admissible Ram window family φ_θ :

1. Prove the explicit-formula support, smoothness, tail, and prime-side trace estimates symbolically. Record the baseline $H(\theta)$ and every error uniformly in T .
2. Derive the normalized atom overlap

$$k_\theta(x) = \frac{\int |\varphi_\theta(t)|^2 e^{-2\pi i x t} dt}{\int |\varphi_\theta(t)|^2 dt}, \quad w_\theta = |k_\theta|^2.$$

Treat removable singularities analytically.

3. Search numerically over q, α , and gap vectors to propose a rational β with a safety margin. Numerical optimization is discovery only.
4. Prove the proposed certificate with outward-rounded interval arithmetic or a proof-assistant-checkable sum-of-squares certificate. Fail on every unresolved cell.
5. Apply Theorem 9.1; choose m using Corollary 9.2; retain exact rational arithmetic until the final decimal display.
6. Independently reproduce the verifier, archive hashes and environment, and separate the finite certificate from the analytic theorem on which it depends.

13.1 Pseudocode for the finite certificate

```
def certify_box(box):
    lower = interval_lower_bound(F_q, box)
    if lower >= beta:
        return VERIFIED
    tangent = convex_tangent_lower_bound(F_q, box)
    if tangent is not None and tangent >= beta:
        return VERIFIED
    if box_is_terminal(box):
        return FAILURE
    left, right = bisect_widest_coordinate(box)
    return certify_box(left) and certify_box(right)
```

The implementation must use directed rounding in the favorable direction for every lower bound. Cached tables require content hashes, and a certificate must state the exact target, grid, precision, source commit, dependency versions, and terminal policy.

13.2 A global-positivity branch

The density pipeline should run in parallel with a stronger branch:

1. choose nested Ram atom spaces dense in the admissible Weil space;
2. evaluate every Weil Gram entry from the prime side with certified errors;
3. obtain lower eigenvalue bounds with interval linear algebra;
4. prove approximation bounds such as (72);
5. search for a uniform margin or a structural factorization of every finite Gram matrix.

A negative certified principal minor would identify a test direction that must be analyzed carefully; it would not by itself be a certified off-line zero unless the full explicit-formula and admissibility framework converts it through Weil’s criterion.

14 No Present Route to a Disproof

To disprove RH one must exhibit and rigorously certify a nontrivial zero with $\operatorname{Re} \rho \neq 1/2$. A numerical candidate would require, at minimum:

1. an interval rectangle wholly off the critical line;
2. a rigorous argument-principle or equivalent zero-count enclosure;
3. exclusion of poles and boundary zeros;
4. sufficient precision to control ζ or ξ throughout the boundary;
5. reproducible independent verification.

Ram transforms could prioritize suspicious regions, precondition oscillatory integrals, or combine multiple local diagnostics. They cannot turn a floating-point near-zero into a theorem. Neither the supplied paper, the prior Ram corpus, nor the present computations produced an off-line zero.

15 Advantages, Limitations, and Comparison

Aspect	Ram–Gram approach	Classical/source approach
Representation	Flexible window, panel, scale, and derivative atoms; explicit Gram geometry.	A carefully optimized test family tailored to the explicit formula.
Transparency	Separates atom design, overlap, finite certificate, and global lifting.	Shorter once the optimized family and trace theorem are accepted.
Computation	Natural interval branch-and-bound, multiwindow experiments, and conditioning diagnostics.	Already supplies a compact reproducible seven-gap verifier.
Arithmetic burden	Every new atom must preserve prime-side support and admit uniform trace estimates.	Those estimates are available for the specific optimized family in the companion preprint.

Aspect	Ram–Gram approach	Classical/source approach
Density improvement	Theorem 9.1 gives an explicit modular formula and optimization criterion.	Gives the concrete 67.3008527% bound.
Path to RH	Dense Ram–Weil positivity is conceptually exact.	The finite trace/inertia method remains insensitive to $o(N)$ off-line zeros.
Main risk	Mistaking a representation, numerical residual, or finite section for a global theorem.	Dependence on very recent analytic and formal artifacts that still warrant independent review.

The principal advantage is modularity: the same proof skeleton can compare windows and certificates while exposing every assumption. The principal disadvantage is equally clear: Ram flexibility enlarges the search space, but each added degree of freedom creates new support, approximation, and conditioning obligations.

16 Future Research Milestones

The following sequence orders work by mathematical value and falsifiability.

1. **Independent analytic audit.** Re-derive the companion preprint’s zero decomposition, trace estimates, endpoint removal, and optimized constant without relying on its formal artifact. Compare the paper and Lean theorem statements symbol for symbol.
2. **Second verifier.** Reimplement the seven-gap certificate in Julia/ArbNumerics, C++/Arb, or Lean interval arithmetic. Require agreement on the exact rational target and kernel-table enclosures.
3. **Certified certificate optimization.** Use global floating-point search only to locate candidate minimizers, then produce rational targets with interval margins. Explore $q = 8, \dots, 12$ before dimension makes naive subdivision prohibitive.
4. **Window co-design.** Parameterize smooth compactly supported windows whose autocorrelations are analytically explicit. Optimize the global bound (76), not the baseline or local certificate separately.
5. **Panel theorem.** Prove one multiwindow rank–trace inequality with a correctly normalized direct-sum Gram matrix and a matching prime-side trace theorem.
6. **Mesoscopic theorem.** Let $q(T) \rightarrow \infty$ slowly and quantify all uniform errors. Even a weak result here would begin to address the fixed-local-statistics barrier.
7. **Ram–Weil approximation theorem.** Construct a nested admissible Ram frame and prove density plus an explicit form-continuity estimate.
8. **Uniform positivity or counterdirection.** Seek either a factorization proving every Weil Gram block PSD or a certified sequence of near-negative directions that reveals exactly where coercivity is lost.
9. **Formal integration.** Formalize Lemma 5.1, Theorem 9.1, and Theorem 11.1; connect a machine-checkable local certificate to the companion explicit-formula theorem.

17 Conclusion

The August 2026 result changes the most promising Ram Calculus direction for RH. The old objective of reading zeta zeros directly as eigenvalues of a bounded FOIRT multiplier should remain retired. The

productive object is instead a finite Ram–Weil Gram matrix whose entries preserve the geometry of simple critical-line zeros and whose prime-side traces remain accessible.

The supplied paper shows that the Gram geometry contains usable information beyond rank and two traces. Its seven-point interval certificate forces a positive convex defect and improves the companion preprint’s 0.6725007... baseline to 0.6730085.... The independent rerun in this report supports the finite computational component, and the strengthened certificate 191/50000 raises the resulting bound to 0.6730213619501666... under the same analytic hypothesis. The general lifting theorem makes this mechanism reusable for other admissible Ram windows.

The investigation also identifies the boundary with unusual clarity. Fixed local statistics cannot see a zero-density exceptional set, while RH forbids even one off-line zero. A complete Ram approach must therefore culminate in uniform global positivity: every finite Ram–Weil block, a dense admissible limit, and an approximation margin strong enough to preserve positivity. That theorem is not presently available. The new density method is genuine progress and an excellent platform for further certified work, but it is not a proof or disproof of RH.

A Exact Arithmetic for the Source Corollary

For $q = 7$, $\alpha = 1/3000$, $\beta = 19/5000$, and $m = 269$,

$$C = 6\alpha = \frac{1}{500}, \quad (80)$$

$$A_m = \frac{19}{5000}(269 - 6) = \frac{4997}{5000}, \quad (81)$$

$$m - A_m = 269 - \frac{4997}{5000} = \frac{1,340,003}{5000}, \quad (82)$$

$$mH_{\text{MT}} - C(m - 1) = 269H_{\text{MT}} - \frac{268}{500} = \frac{1,345,000H_{\text{MT}} - 2680}{5000}. \quad (83)$$

Their quotient is exactly the rational-affine expression in (1). Keeping these quantities rational until the final evaluation prevents a displayed decimal from becoming an input to the proof.

B TR220 Strengthened Certificate and Exact Arithmetic

The stronger run used

$$\beta = \frac{191}{50000} = 0.00382, \quad \frac{\beta}{\alpha} = 11.46, \quad 11.46 \times 4000 = 45,840 \quad (84)$$

for the pressure cutoff. Its complete summary was:

```
source_commit=040c5e899e658aed7b56a2a87f501798fe10761d
certificate=seven-point
verified=true
target=F6 >= 191/50000
grid=4000
precision_bits=128
kernel_table_sha256=f79a9147ffc37692b84330e98422fb94cbf601c51d4f9f8ae749982f23838571
initial_boxes=729
nodes=786215
pruned=393472
```

```
splits=392743
maximum_depth=43
interval_pruned=269746
pressure_pruned=2901
second_derivative_table_sha256=
    cc98102590dba6e1a982a5a4c9fcd755848a93483db67edd6634813a35e5e3f5
surviving_gap_components_cells=[3808,4780];[7218,9371];[10563,44917]
surviving_gap_components_count=3
tangent_pruned=120825
```

For comparison, the official 19/5000 rerun produced

```
kernel_table_sha256=a9992300d2bf71665aa2b6bd2727e798624cd297103bb200c7f0ca2baea55a2c
second_derivative_table_sha256=7913
    c5511a572c32dd573cd53123d8cf3ddf73d3ec63b1aa823faae2ae83570a
```

Elapsed time is omitted from the mathematical certificate because it is machine dependent; the audited local run took approximately 246 seconds.

For $m = 267$,

$$C = \frac{1}{500}, \quad (85)$$

$$A_m = \frac{191}{50000}(267 - 6) = \frac{49851}{50000}, \quad (86)$$

$$m - A_m = \frac{13,300,149}{50000}, \quad (87)$$

$$mH_{\text{MT}} - C(m - 1) = 267H_{\text{MT}} - \frac{266}{500} = \frac{13,350,000H_{\text{MT}} - 26,600}{50000}. \quad (88)$$

The quotient is (2). The standalone wrapper and text record distributed with TR220 are `verify_tr220_seven.py` and `TR220-seven-point-certificate.txt`.

C Adversarial Review Checklist

Before any future TR claims an RH consequence, verify each item below.

1. Are zero counts with multiplicity distinguished from distinct and simple zeros?
2. Are off-line zeros retained as functional-equation pairs rather than silently projected onto the line?
3. Is every transform identity valid in a specified function space and on a stated operator domain?
4. If eigenvalues are claimed, are there normalizable eigenvectors and an exact operator realization?
5. If compact resolvent is claimed, is the operator unbounded on an infinite-dimensional space with a justified domain?
6. Are finite-section convergence and spectral pollution controlled?
7. Are explicit-formula test functions admissible and are prime-side sums convergent or correctly regularized?
8. Are asymptotic errors uniform over the growing family being used?
9. Is a numerical inequality converted to an outward-rounded certificate with an explicit trust base?
10. Does the conclusion exclude a zero-density exceptional set, or does it only improve a proportion?
11. If Weil positivity is invoked, is positivity proved for a dense test space with a continuous limiting form?
12. Can an independent implementation reproduce the exact rational constant and certificate hash?

D Reproducibility Commands

At the audited commit, the repository’s standard workflow is:

```
python -m venv .venv
.venv\Scripts\python -m pip install -e .
.venv\Scripts\python -m unittest discover -s tests -v
.venv\Scripts\zeta-zero-verify seven
```

On non-Windows systems the activation and executable paths differ. The certificate should be compared by fields and hashes rather than elapsed time, which is hardware dependent.

E Selected Ram Corpus References

The RH audit relied on the following local reports in addition to the external literature. The file titles, rather than earlier headline claims, determine their use here.

- TR20, *Ram Transform proof attempt for RH*; TR23, its gap analysis; and TR65, the comprehensive FOIRT proof attempt.
- TR73 and TR96, adelic RamCS/RMT constructions.
- TR94, *Ram Transform Family Paths Toward the Riemann Hypothesis*; TR95 on GRH; and TR97, the corrected conditional formulation.
- TR147, the comprehensive RMT report; TR148, the corrected spectral and zero-density reassessment; TR165, the analytic-core/learned-residual umbrella; and TR184, the RMT roadmap.
- TR207–TR219, the robust, multidimensional, panel, discrete, nonlinear, coupled, microlocal, fractional, resolvent, and domain-instantiation extensions.

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